

Add and Subtract Decimals

Lesson 1-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

3.75 means 3 ones + 7 tenths + 5 hundredths — it is between 3 and 4

Decimal

A number with a dot that shows a part between whole numbers.

In 4.38: the 4 = 4 ones, the 3 = 3 tenths (0.3), the 8 = 8 hundredths (0.08)

Place value

What a digit is worth based on where it sits in a number.

0.7 = $7/10$ — like 7 slices of a pizza cut into 10 equal slices

Tenths

The first spot after the decimal point. It shows parts out of 10.

0.05 = $5/100$ — like 5 pennies out of a dollar (100 pennies)

Hundredths

The second spot after the decimal point. It shows parts out of 100.

15.7 becomes 15.70 so you can subtract $15.70 - 8.36$ with aligned place values

Annex zeros

Adding zeros at the end of a decimal so numbers line up evenly.

Key Ideas & Notes

- The space station's main fuel tank reads 128.75 liters.
- A resupply shuttle delivers 46.8 liters of fuel, but the thrusters used 19.35 liters during docking.
- What is the fuel level after both events?
- Plot each decimal on the number line, then find the sum or difference as indicated.

Think About It

- What is the starting fuel amount?
- Which operation adds fuel and which removes fuel?
- Why is it important to line up the decimal points?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The fuel tank reads 128.75 liters, a shuttle delivers 46.8 liters, and docking thrusters used 19.35 liters. Why must you line up the decimal points before adding or subtracting these amounts?

Level 1 support

Start here: Lining up the decimal points keeps each place value matched.

I line up the ____ so tenths add to tenths and hundredths add to hundredths.

Alineo los ____ para que los décimos se sumen con décimos y los centésimos con centésimos.

If the points are not aligned, then ____.

Si los puntos no están alineados, entonces ____.

Word bank: decimal place value tenths hundredths annex zeros

Listen for: Students explain that aligning decimal points lines up matching place values (tenths under tenths, hundredths under hundredths) so the addition and subtraction are correct.

Level 2

Estimate the final fuel level by rounding each number, then explain how that estimate checks your exact answer of $128.75 + 46.8 - 19.35$.

- I estimate about ____ liters because I round to ____.
- My exact answer is reasonable because it is close to ____.

EXPLORE

When you added 128.75 and 46.8, how did you handle 46.8 having fewer decimal places, and when did you need to regroup?

Level 1 support

Start here: You can annex a zero so 46.8 becomes 46.80 and every place value lines up.

I annexed a zero to write 46.8 as ____ so the place values matched.

Agregué un cero para escribir 46.8 como ____ para que los valores posicionales coincidieran.

I had to regroup when ____.

Tuve que reagrupar cuando ____.

Word bank: decimal place value tenths hundredths annex zeros

Listen for: Students annex a zero to write 46.80, aligning hundredths, and explain regrouping happens when a column sum is 10 or more (or borrowing is needed when subtracting).

Level 2

Why does annexing a zero to make 46.8 into 46.80 not change its value? Justify with place value.

- Annexing a zero does not change the value because ____.
- The extra zero is in the ____ place, which means ____.

CONNECT

A runner's three laps are 2.45, 2.6, and 2.38 minutes. How do you add decimals with different numbers of places to find her total time?

Level 1 support

Start here: Annex zeros so all times have the same places, then line up and add.

Her total time is ____ minutes because I added ____, ____, and ____.

Su tiempo total es ____ minutos porque sumé ____, ____ y ____.

I wrote 2.6 as ____ so every place value lined up.

Escribí 2.6 como ____ para que cada valor posicional se alineara.

Word bank: (decimal) (place value) (tenths) (hundredths) (annex zeros)

Listen for: Students annex zeros (2.60), align the decimal points, and add $2.45 + 2.60 + 2.38 = 7.43$ minutes.

Level 2

Her personal record is 7.5 minutes. Did she beat it? Compare 7.43 and 7.5 using place value and justify.

- She ____ beat her record because 7.43 is ____ than 7.5.
- Comparing the tenths place, ____, so ____.

REFLECT

Why does lining up place values matter MORE with decimals than with whole numbers, where you can just line up the right edge?

Level 1 support

Start here: Decimals have different lengths, so only the decimal point guarantees matched places.

With decimals I must line up the _____, not just the right edge, because _____.

Con decimales debo alinear los _____, no solo el borde derecho, porque _____.

If I lined up the right edge, then _____.

Si alineara el borde derecho, entonces _____.

Word bank: decimal place value tenths hundredths annex zeros

Listen for: Students explain that decimals can have different numbers of places, so lining up the decimal point (not the right edge) is what keeps tenths with tenths and hundredths with hundredths.

Level 2

Create an addition where lining up the right edge instead of the decimal point gives a wrong answer, and show the correct sum.

- If I lined up 3.5 and 0.42 by the right edge, I would wrongly get _____.
- The correct sum is _____ because _____.

Guided Examples

Example 1

What is $14.6 + 3.85$?

Solution: Line up decimals: $14.60 + 3.85$. Hundredths: $0 + 5 = 5$. Tenths: $6 + 8 = 14$, write 4 carry 1. Ones: $4 + 3 + 1 = 8$. Tens: 1. Answer: 18.45.

Answer: A. 18.45

Example 2

What is $20.3 - 7.48$?

Solution: Rewrite as $20.30 - 7.48$. Annex a zero so both have two decimal places. Work carefully with borrowing: $20.30 - 7.48 = 12.82$.

Answer: A. 12.82

Example 3

What is $9.05 + 4.7$?

Solution: Line up decimals: $9.05 + 4.70$. Hundredths: $5 + 0 = 5$. Tenths: $0 + 7 = 7$. Ones: $9 + 4 = 13$. Answer: 13.75.

Answer: A. 13.75

Write About the Math

The Writing Revolution

I can explain my steps using the words decimal, place value, tenths, hundredths, and annex zeros.

1. Kernel Sentence subject + verb

Model: Decimal is a number with a dot that shows a part between whole numbers.

Decimal es un número con un punto que muestra una parte entre números enteros.

Write a kernel sentence about decimal. Use a subject and a verb.

Escribe una oración base sobre decimal. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Decimal matters in math

Decimal importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Decimal matters in math because ____.

Decimal importa en matemáticas porque ____.

but
pero

Decimal matters in math, but ____.

Decimal importa en matemáticas, pero ____.

so
entonces

Decimal matters in math, so ____.

Decimal importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about decimal.
Di un hecho verdadero sobre decimal.

Decimal ____.

Question *Pregunta*

Ask a question about decimal.
Haz una pregunta sobre decimal.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about decimal.
Muestra entusiasmo sobre decimal.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with decimal.
Dile a un compañero qué hacer con decimal.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I line up the ____ so tenths add to tenths and hundredths add to hundredths.

Alineo los ____ para que los décimos se sumen con décimos y los centésimos con centésimos.

If the points are not aligned, then ____.

Si los puntos no están alineados, entonces ____.

I annexed a zero to write 46.8 as ____ so the place values matched.

Agregué un cero para escribir 46.8 como ____ para que los valores posicionales coincidieran.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $50 - 23.47$?

- A. 26.53
- B. 27.53
- C. 26.63
- D. 27.47

Show your work:

2. A student buys items costing \$4.59, \$12.75, and \$3.80. What is the total cost?

- A. \$21.14
- B. \$20.14
- C. \$21.04
- D. \$21.24

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A runner's lap times are 2.45 minutes, 2.6 minutes, and 2.38 minutes. Her personal record is 7.5 minutes total. Did she beat her record? Show all your work with decimal alignment and explain.

Sentence starter: First I lined up the decimals: $2.45 + 2.60 + 2.38 = \underline{\quad}$. Compared to 7.50, the runner beat her record because .

Show your work:

Reflect — Exit Ticket

What is $32.6 - 14.85$?

- A. 17.75
- B. 18.75
- C. 17.85
- D. 17.25

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 26.53 — Rewrite 50 as 50.00. Then $50.00 - 23.47 = 26.53$. You must borrow across multiple places.
2. **Try It 2:** A. $\$21.14 - 4.59 + 12.75 + 3.80 = 21.14$. Line up decimal points and add column by column.
3. **Exit Ticket:** A. 17.75 — Rewrite as $32.60 - 14.85$. Hundredths: $0 - 5 \rightarrow$ borrow $\rightarrow 10 - 5 = 5$. Tenths: $5 - 8 \rightarrow$ borrow $\rightarrow 15 - 8 = 7$. Ones: $1 - 4 \rightarrow$ borrow $\rightarrow 11 - 4 = 7$. Tens: $2 - 1 = 1$. Answer: 17.75.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Decimal is a number with a dot that shows a part between whole numbers.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).